ESXi Performance

Workflow and Troubleshooting

ESXTOP OVERVIEW:

- Esxtop allows monitoring and collection of data for all system resources: CPU, memory, disk and network. When used interactively, this data can be viewed on different types of screens; one each for CPU statistics, memory statistics, network statistics and disk adapter statistics.

esxtop is the primary real-time performance monitoring tool for vSphere:

- This tool can be run from a host's local vSphere ESXi Shell as esxtop.
- This tool can also be run remotely from VMware vSphere® Command-Line Interface as resxtop.
- esxtop provides much of the same information as vSphere Client but supplements the output with more detailed information.
- esxtop works like the top performance utility found in Linux operating systems.
- esxtop displays information in percentages of sample time (5 seconds by default)

About Esxtop

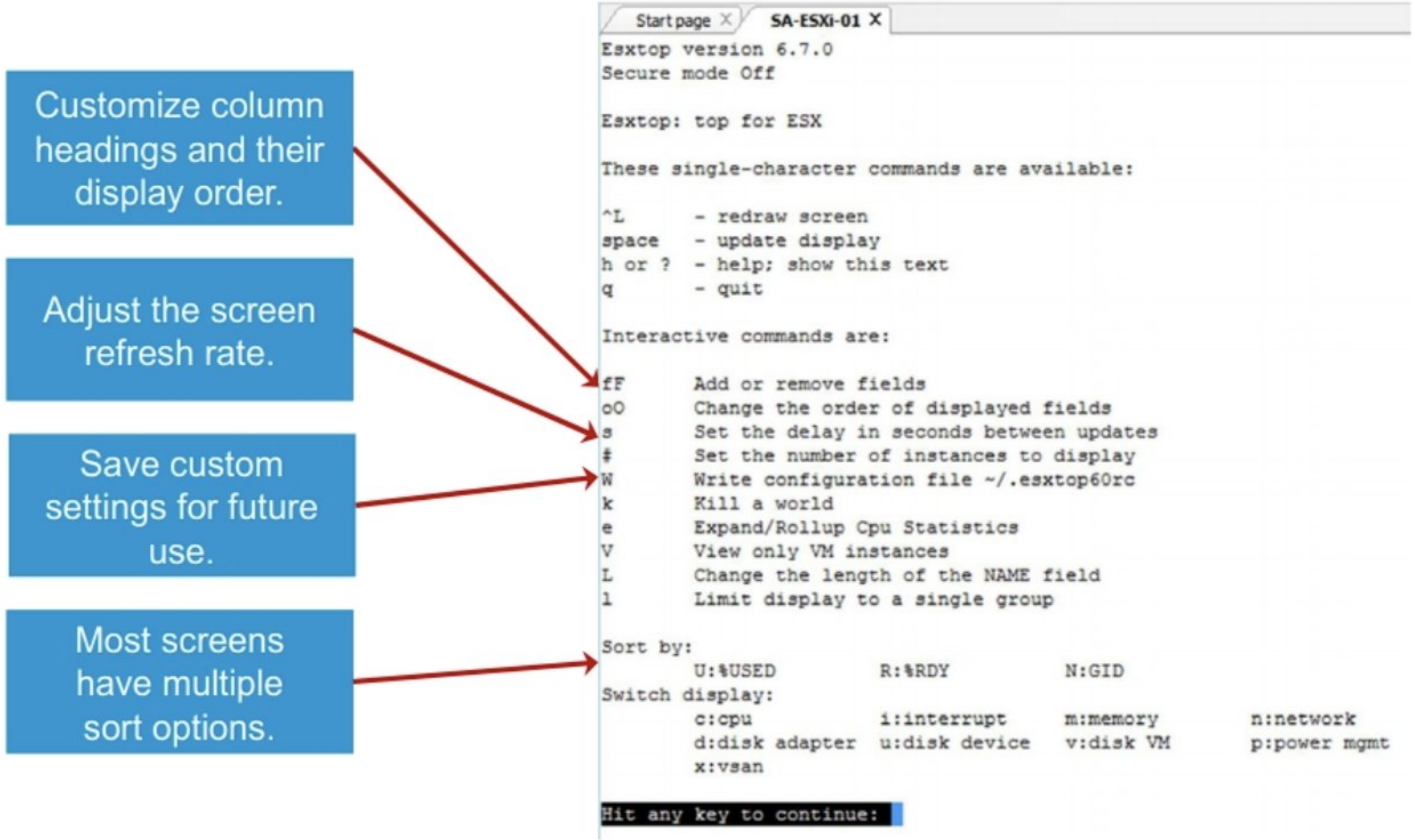
You view key performance indicators on individual resource screens by entering the appropriate keys. Commands are case-sensitive

You use lowercase and uppercase letters to specify which fields appear in which order on the CPU, memory, storage adapter, storage device, virtual machine storage, and network panels. You can add or remove columns containing data in the respective esxtop panels

c	CPU screen (default)
m	Memory screen
d	Disk (adapter) screen
u	Disk (device) screen
v	Virtual disk view (lowercase v)
n	Network screen
f/F	Add or remove statistic columns
V	Virtual machine view (uppercase V)
h	Help
q	Quit

Getting Help in esxtop

The `esxtop` help utility lists the available commands. Enter **h** or **?** to display the help screen.



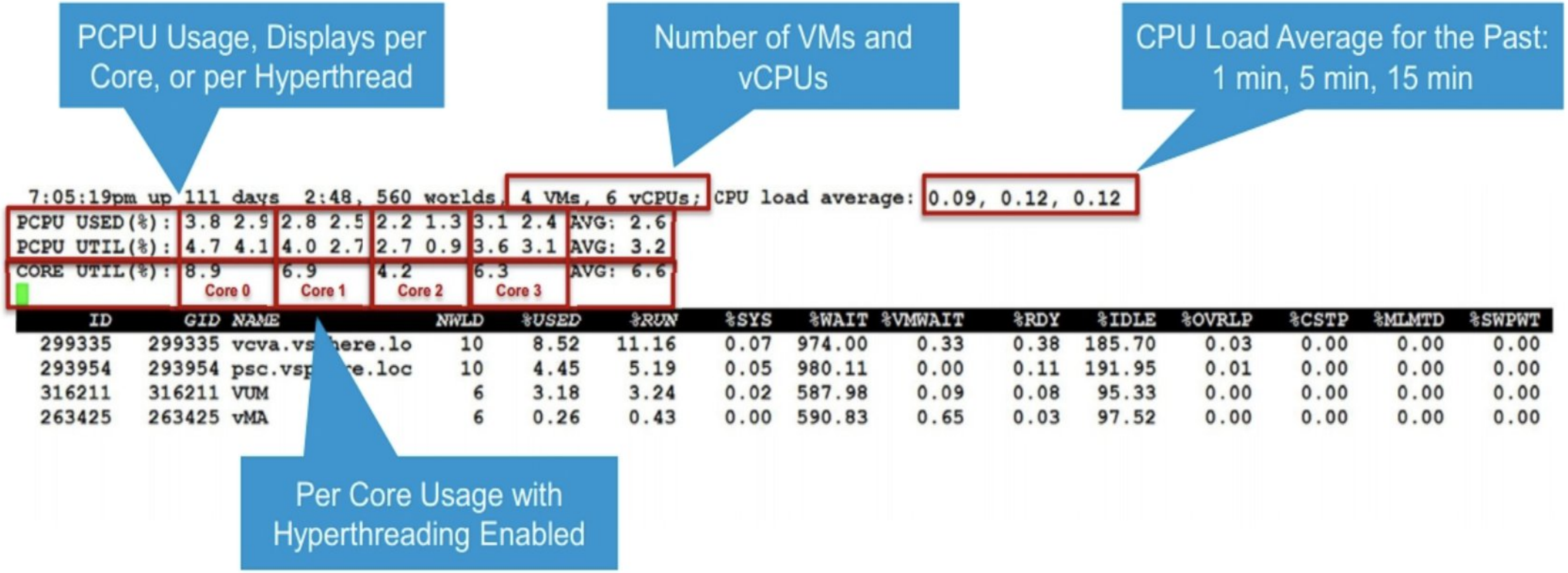
World ID and GID

Esxtop uses worlds and groups as the entities to show CPU usage. A **world** is an ESX Server VMkernel schedulable entity, similar to a process or thread in other operating systems. A **group** contains multiple worlds.

In esxtop, a group is a resource pool, a running VM, or a non-VM world. For worlds belonging to a VM, statistics for the running VM are shown and are identified by the VM's group ID (GID).

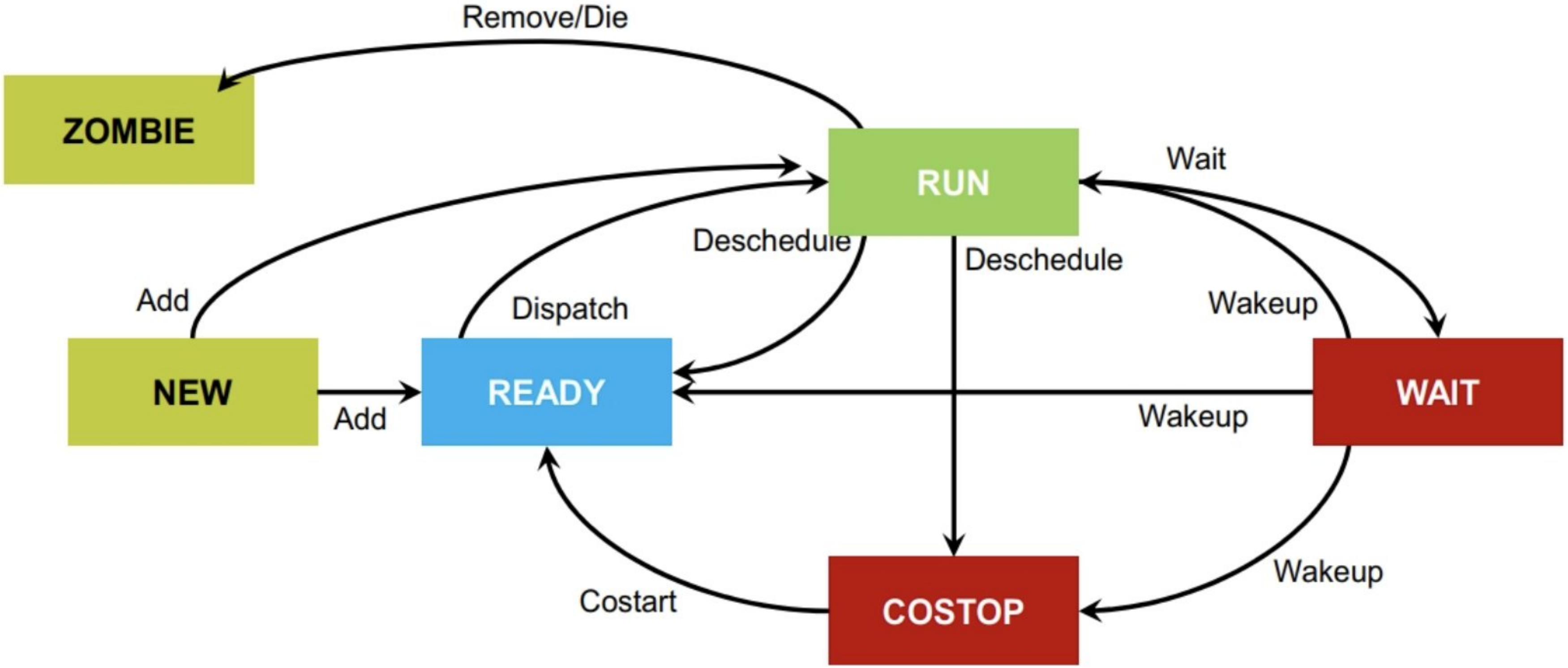
Using esxtop to Monitor ESXi Host CPU

Enter **c** to display CPU information about the ESXi host.



Transitioning Between World States

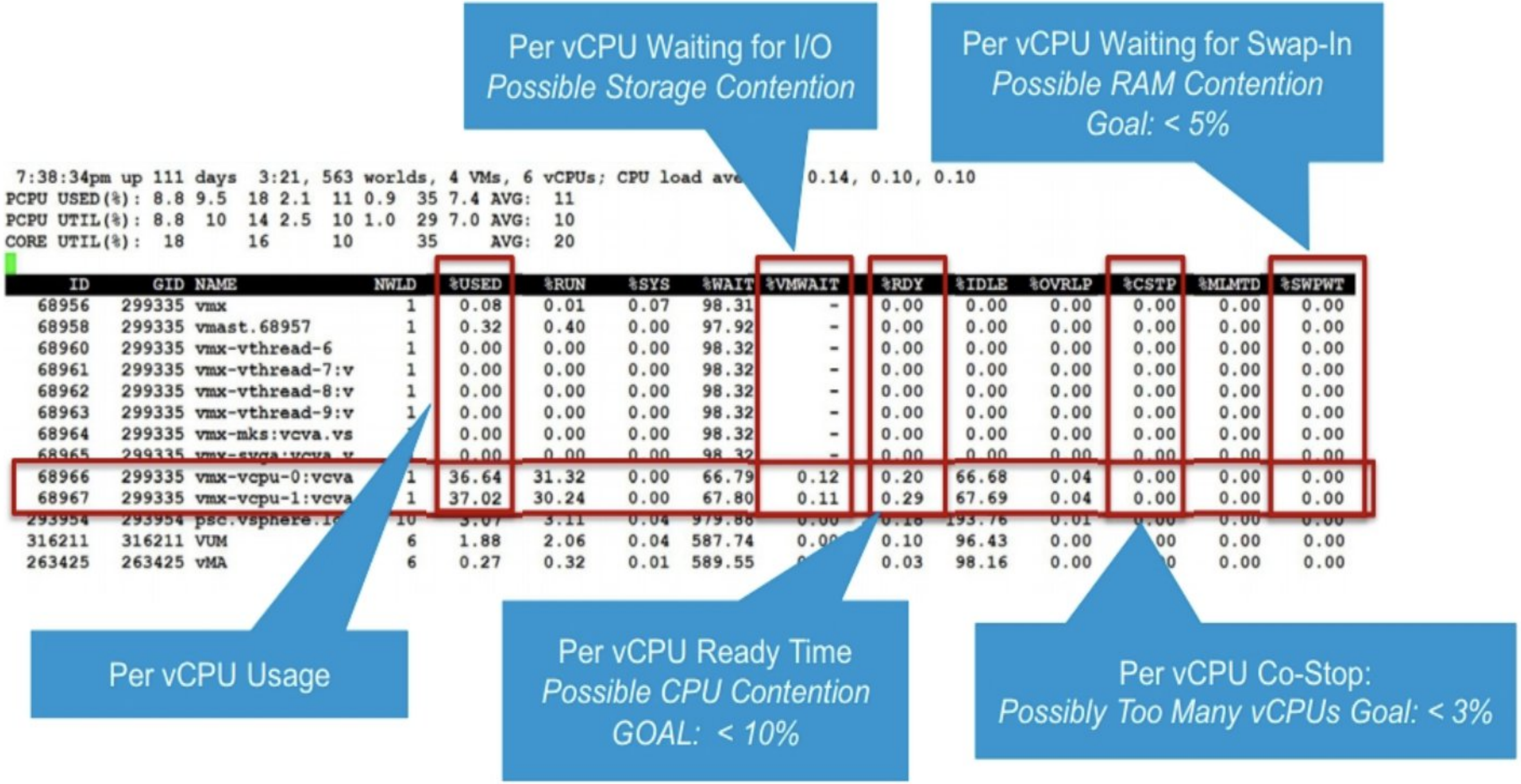
A world transitions through various states. Initially, a world is associated with the run state or the ready state.



Use the following CPU key performance indicators for VMs

Key Performance Indicator	esxtop Counter
High vCPU usage	%USED
Low vCPU usage	%USED
Ready time	%RDY
Co-stop time	%CSTP
Max limited time	%MLMTD
VM wait time	%VMWAIT
Swap wait time	%SWPWT
Number of members in a VM world	NWLD

Using esxtop to Monitor Virtual Machine vCPU



Important Metrics to Monitor

- Monitoring both high-usage values and ready time is important to accurately depict CPU performance.

High-usage values alone do not always indicate poor performance:

• This value is often an indicator of high usage.

• This value is a goal of many organizations.

Ready time reflects the idea of queuing in the CPU scheduler:

• Ready time is the best indicator of possible CPU performance problems.

• Ready time occurs when more CPU resources are being requested by virtual machines than can be scheduled on the given physical CPUs.

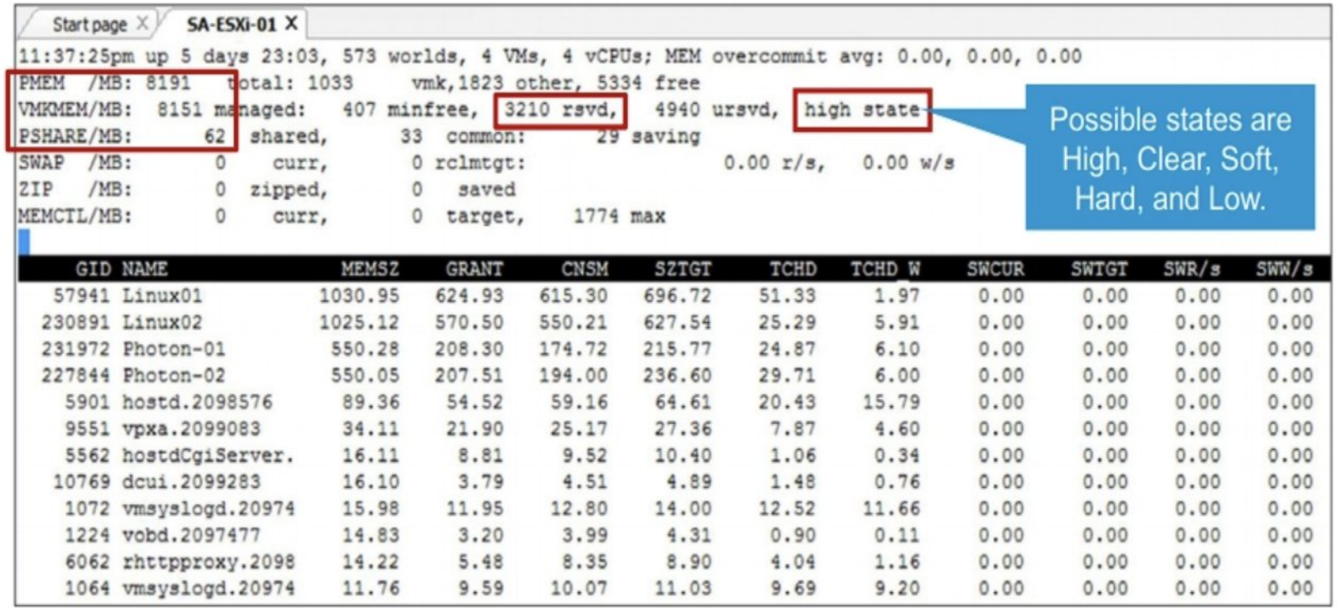
continue

- High usage is not necessarily an indication of poor CPU performance, although it is often mistakenly understood as such. Administrators in the physical space view high usage as a warning sign, due to the limitations of physical architectures. However, one of the goals of virtualization is to efficiently use the available resources. Detecting high CPU usage, along with queuing, might indicate a performance problem.
- As an analogy, consider a freeway. If every lane in the freeway is packed with cars but traffic is moving at a reasonable speed, you have high use but good performance. The freeway is doing what is was designed to do: moving a large number of vehicles at its maximum capacity.
- Now, imagine the same freeway at rush hour. More cars than the maximum capacity of the freeway are trying to share the limited space. Traffic speed reduces to the point where cars line up (queue) at the on-ramps while they wait for shared space to open up so that they can merge onto the freeway. This condition indicates high use with queuing and that a performance problem exists.
- Ready time reflects the idea of queuing in the CPU scheduler. Ready time is the amount of time that a vCPU was ready to run but was asked to wait due to an overcommitted physical CPU.

Using esxtop to Monitor Memory Usage

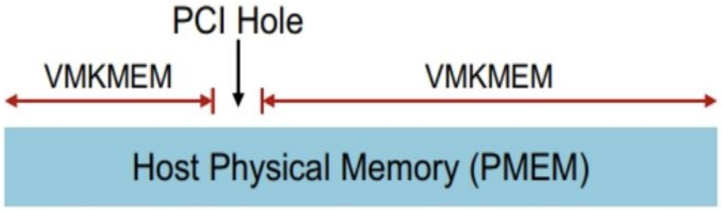
`esxtop` offers several options for monitoring memory usage.

Enter **`m`** to display the memory screen.




```
11:37:25pm up 5 days 23:03, 573 worlds, 4 VMs, 4 vCPUs; MEM overcommit avg: 0.00, 0.00, 0.00
PMEM   /MB: 8191    total: 1033     vmk,1823 other, 5334 free
VMKMEM/MB:  8151 managed:   407 minfree,  3210 rsvd,   4940 ursvd,  high state
PSHARE/MB:        62  shared,      33  common:       29 saving
SWAP   /MB:        0    curr,       0 rclmtgt:                  0.00 r/s,   0.00 w/s
ZIP    /MB:        0  zipped,       0   saved
MEMCTL/MB:         0    curr,       0  target,     1774 max
```

GID	NAME	MEMSZ	GRANT	CNSM	SZTGT	TCHD	TCHD_W	SWCUR	SWTGT	SWR/s	SWW/s
57941	Linux01	1030.95	624.93	615.30	696.72	51.33	1.97	0.00	0.00	0.00	0.00
230891	Linux02	1025.12	570.50	550.21	627.54	25.29	5.91	0.00	0.00	0.00	0.00
231972	Photon-01	550.28	208.30	174.72	215.77	24.87	6.10	0.00	0.00	0.00	0.00
227844	Photon-02	550.05	207.51	194.00	236.60	29.71	6.00	0.00	0.00	0.00	0.00
5901	hostd.2098576	89.36	54.52	59.16	64.61	20.43	15.79	0.00	0.00	0.00	0.00
9551	vpxa.2099083	34.11	21.90	25.17	27.36	7.87	4.60	0.00	0.00	0.00	0.00
5562	hostdCgiServer.	16.11	8.81	9.52	10.40	1.06	0.34	0.00	0.00	0.00	0.00
10769	dcui.2099283	16.10	3.79	4.51	4.89	1.48	0.76	0.00	0.00	0.00	0.00
1072	vmsyslogd.20974	15.98	11.95	12.80	14.00	12.52	11.66	0.00	0.00	0.00	0.00
1224	vobd.2097477	14.83	3.20	3.99	4.31	0.90	0.11	0.00	0.00	0.00	0.00
6062	rhttpproxy.2098	14.22	5.48	8.35	8.90	4.04	1.16	0.00	0.00	0.00	0.00
1064	vmsyslogd.20974	11.76	9.59	10.07	11.03	9.69	9.20	0.00	0.00	0.00	0.00



VMKMEM: Memory Managed by the VMkernel

Using esxtop to Monitor Memory Usag

- The following metrics are useful in monitoring memory usage:

• PMEM: The total amount of physical memory on your host in megabytes.

• VMKMEM: The memory that is managed by the VMkernel in megabytes.

• PSHARE: The saving field shows you how much memory you saved because of TPS. In the example, two virtual machines are running, with a memory savings of 62 MB.

• Memory state: This value can be high, clear, soft, hard, or low. This value refers to the current state of the VMkernel’s memory. This value also tells you whether the VMkernel has enough free memory to perform its critical operations.

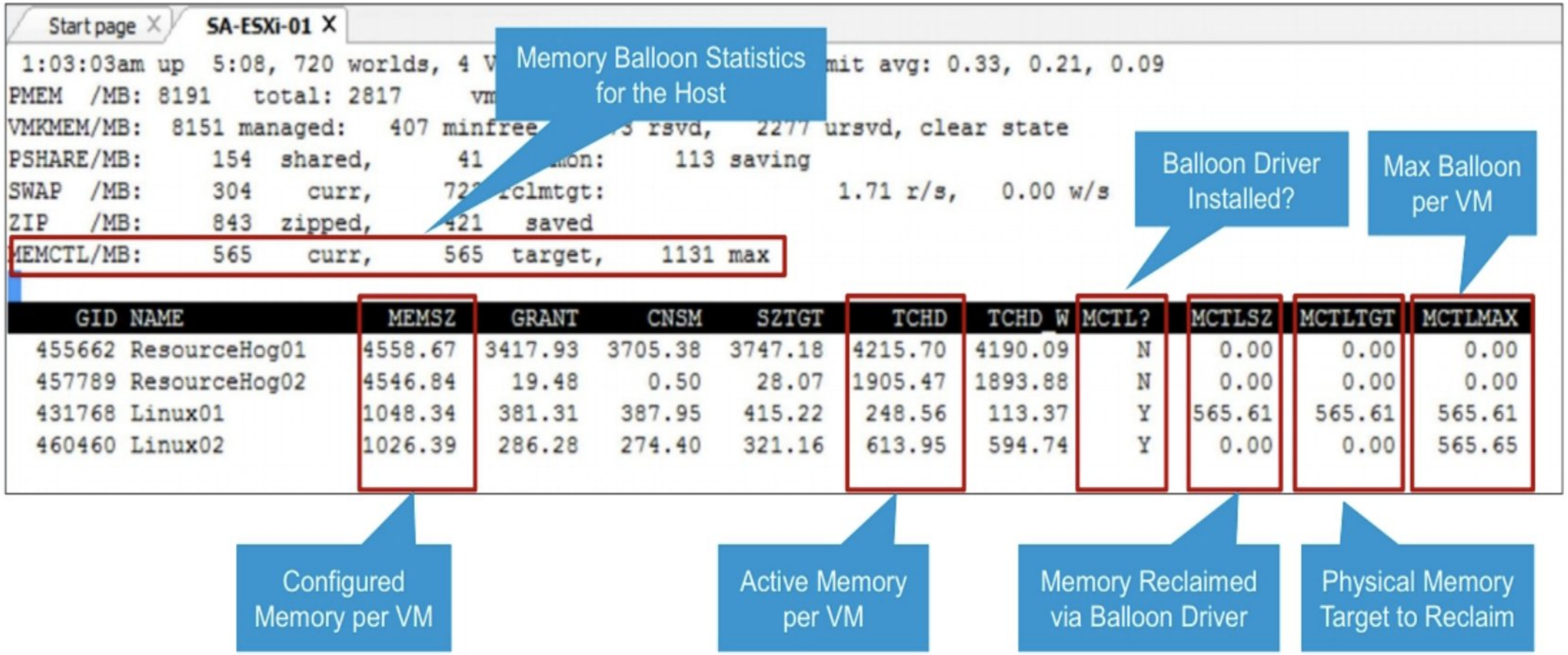
Continue

- If the state is high, the VMkernel has sufficient memory to perform critical tasks. However, if the state is clear, soft, hard, or low, the VMkernel is unable to maintain adequate free memory. Typically, you should see only a state of high, even if you are overcommitting your memory. ESXi employs efficient mechanisms for reserving memory for the VMkernel. However, if your host is severely overcommitted, you might see a state other than high.

• A PCI hole is the memory reserved by the PCI devices on the host. This memory is used solely by the PCI devices for I/O purposes, such as DMA ring buffers. This memory cannot be used by the VMkernel. Reserving memory by PCI devices is common in all operating systems. The amount of memory used for PCI devices is part of the other value, in the PMEM/MB line.

Host Ballooning Activity in esxtop

On the `esxtop` memory screen, enter **f** to display the fields window and then enter **j** to add the memory control fields used to monitor virtual machine ballooning activity.



Continue

The following metrics are useful in monitoring ballooning activity:

• MEMCTL/MB: This line displays the memory balloon statistics for the entire host. All numbers are in megabytes:

• curr is the total amount of physical memory reclaimed using the balloon driver (vmmemctl).

• target is the total amount of physical memory ESXi wants to reclaim with the balloon driver.

• max is the maximum amount of physical memory that the ESXi host can reclaim with the balloon driver.

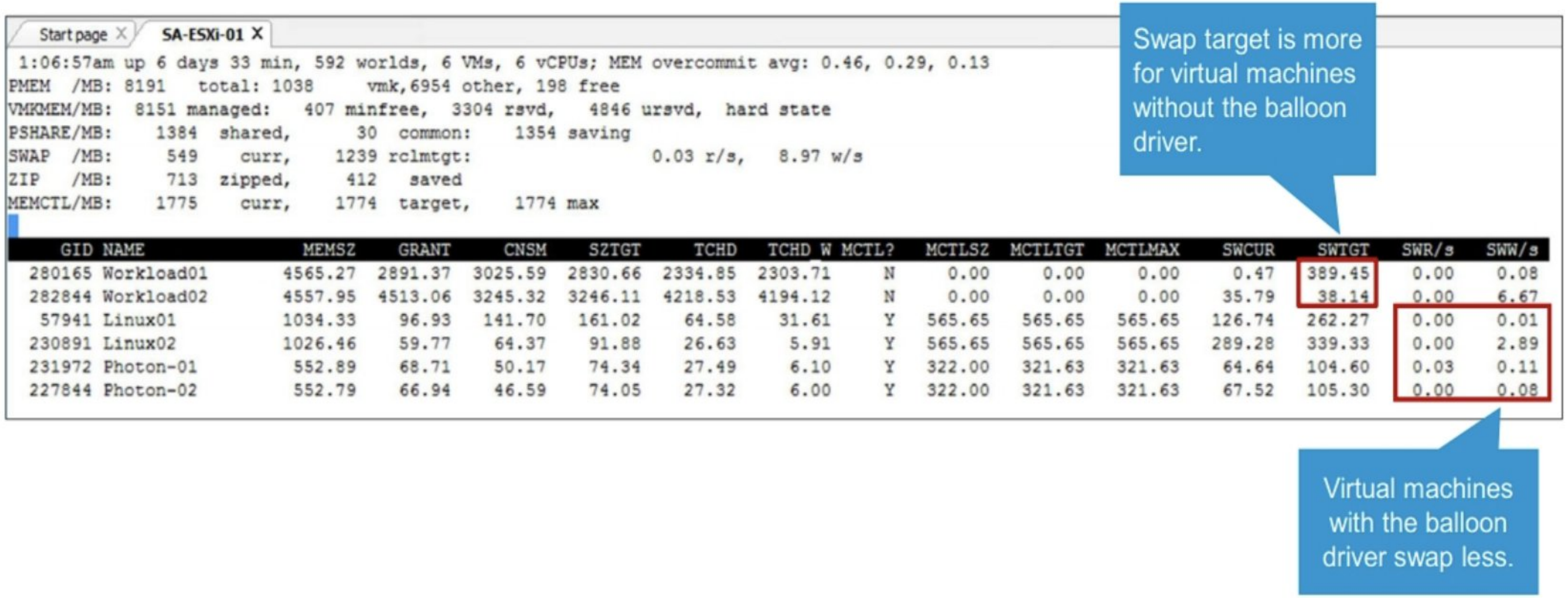
Continue

- MCTL?: This value, which is either Y (for yes) or N (for no), indicates whether the balloon driver is installed in the virtual machine.
- MCTLSZ: This value is reported for each virtual machine and represents the amount of physical memory that the balloon driver is holding for use by other virtual machines.
- MCTLTGT: This value is the amount of physical memory that the host wants to reclaim from the virtual machine through ballooning

Host Ballooning Activity in esxtop

The `esxtop` memory screen shows where ballooning is used and where swapping is used.

Enter **f** to display the fields window and then enter **k** to add the swap statistics fields.



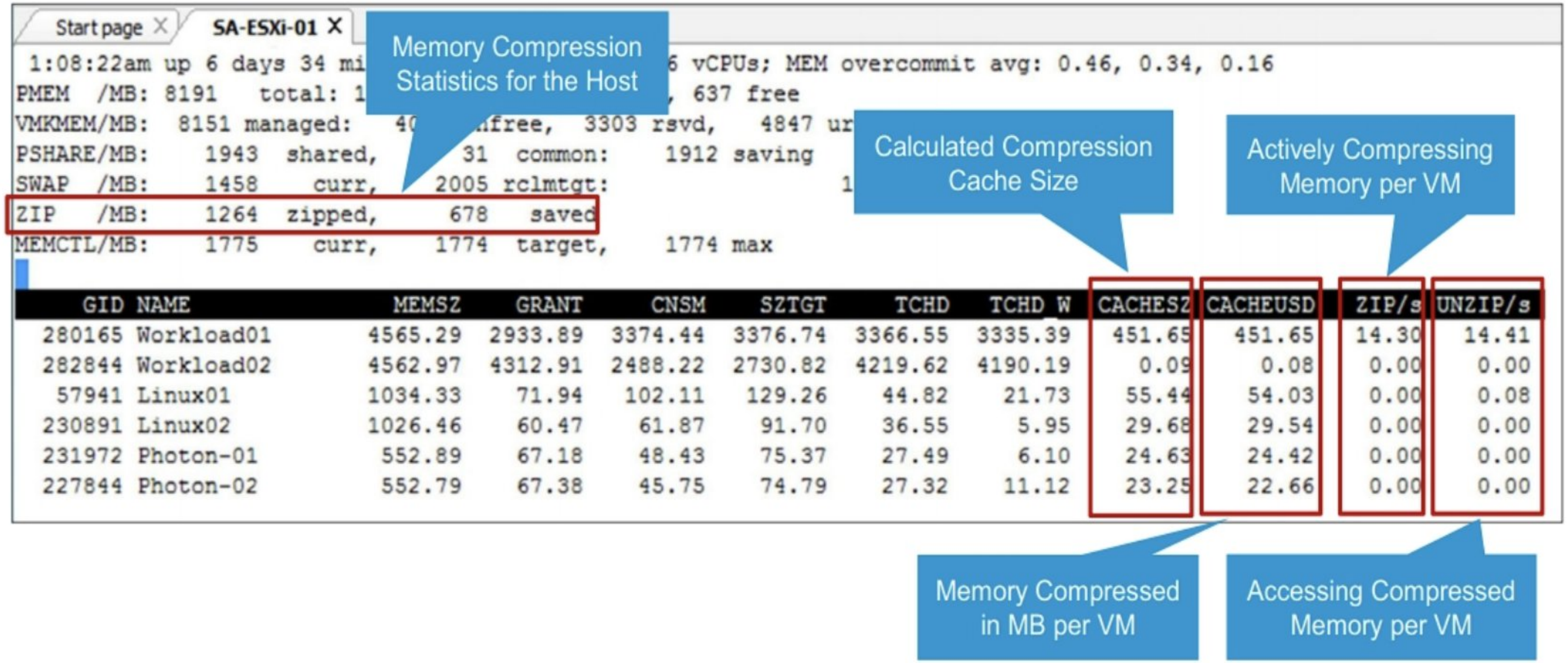

```
 1:06:57am up 6 days 33 min, 592 worlds, 6 VMs, 6 vCPUs; MEM overcommit avg: 0.46, 0.29, 0.13
PMEM  /MB: 8191   total: 1038    vmk,6954 other, 198 free
VMKMEM/MB:  8151 managed:   407 minfree,  3304 rsvd,   4846 ursvd,  hard state
PSHARE/MB:    1384  shared,      30  common:    1354 saving
SWAP  /MB:     549    curr,    1239 rclmtgt:                0.03 r/s,   8.97 w/s
ZIP   /MB:     713  zipped,     412   saved
MEMCTL/MB:    1775    curr,    1774  target,    1774 max

   GID NAME           MEMSZ   GRANT    CNSM   SZTGT    TCHD  TCHD_W MCTL?  MCTLSZ MCTLTGT MCTLMAX   SWCUR   SWTGT   SWR/s   SWW/s
  280165 Workload01  4565.27 2891.37 3025.59 2830.66 2334.85 2303.71     N    0.00    0.00    0.00    0.47  389.45    0.00    0.08
  282844 Workload02  4557.95 4513.06 3245.32 3246.11 4218.53 4194.12     N    0.00    0.00    0.00   35.79   38.14    0.00    6.67
   57941 Linux01     1034.33   96.93  141.70  161.02   64.58   31.61     Y  565.65  565.65  565.65  126.74  262.27    0.00    0.01
  230891 Linux02     1026.46   59.77   64.37   91.88   26.63    5.91     Y  565.65  565.65  565.65  289.28  339.33    0.00    2.89
  231972 Photon-01    552.89   68.71   50.17   74.34   27.49    6.10     Y  322.00  321.63  321.63   64.64  104.60    0.03    0.11
  227844 Photon-02    552.79   66.94   46.59   74.05   27.32    6.00     Y  322.00  321.63  321.63   67.52  105.30    0.00    0.08
```

Continue

- The example shows why ballooning is preferable to swapping. Multiple virtual machines are running, and all the virtual machines are running a memory-intensive application. The MCTL? field shows that the virtual machines named Workload01 and Workload02 do not have the balloon driver installed. The balloon driver is installed on all the other virtual machines. If the balloon driver is not installed, typically VMware Tools was not installed on the virtual machine.
- The virtual machines with no balloon driver installed usually have a higher swap target (SWTGT) than the virtual machines that have the balloon driver installed. Virtual machines that have the balloon driver installed can reclaim memory by using the balloon driver. If ESXi cannot reclaim memory through ballooning, then it must resort to other methods, such as host-level swapping.

Host Memory Compression Activity in esxtop

On the `esxtop` memory screen, enter **f** to display the fields window and then enter **q** to add the memory compression fields to monitor virtual machine compression activity.



Host Memory Compression

- ESXi provides a memory compression cache to improve virtual machine performance when you use memory overcommitment. Memory compression is enabled by default. When a host's memory becomes overcommitted, ESXi compresses virtual pages and stores them in memory.
- Because accessing compressed memory is faster than accessing memory that is swapped to disk, memory compression in ESXi enables you to overcommit memory without significantly hindering performance. When a virtual page needs to be swapped, ESXi first tries to compress the page. Pages that can be compressed to 2 KB or smaller are stored in the virtual machine's compression cache, increasing the capacity of the host.

Continue

- You can monitor the following values in esxtop:

- CACHESZ (MB): Compression memory cache size
- CACHEUSD (MB): Used compression memory cache
- ZIP/s (MB/s): Compressed memory per second
- UNZIP/s (MB/s): Decompressed memory per second

Monitoring Disk Throughput by Storage Adapter

Use the `esxtop` command to monitor disk throughput.

Enter **d** to display disk throughput by storage adapter.

Start page X SA-ESXi-01 X

9:25:08pm up 4 days 20:51, 556 worlds, 2 VMs, 2 vCPUs; CPU load average: 0.05, 0.07, 0.05

ADAPTR	PATH	NPTH	CMDS/s	READS/s	WRITES/s	MBREAD/s	MBWRTN/s	DAVG/cmd	KAVG/cmd	GAVG/cmd	QAVG/cmd
vmhba0	-	1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
vmhba1	-	4	0.38	0.00	0.38	0.00	0.00	0.67	0.03	0.70	0.00
vmhba64	-	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
vmhba65	-	10	0.76	0.00	0.76	0.00	0.00	1.42	0.01	1.43	0.00

Disk throughput appears in these columns:

- READS/s and WRITES/s
- MBREAD/s and MBWRTN/s

Continue

Disk throughput can also be monitored using esxtop:

• READS/s: Number of disk reads per second

• WRITES/s: Number of disk writes per second

The sum of reads per second and writes per second equals IOPS.

IOPS is a common benchmark for storage subsystems and can be measured with tools like Iometer.

If you prefer, you can monitor throughput by using the following metrics instead:

• MBREAD/s: Number of megabytes read per second

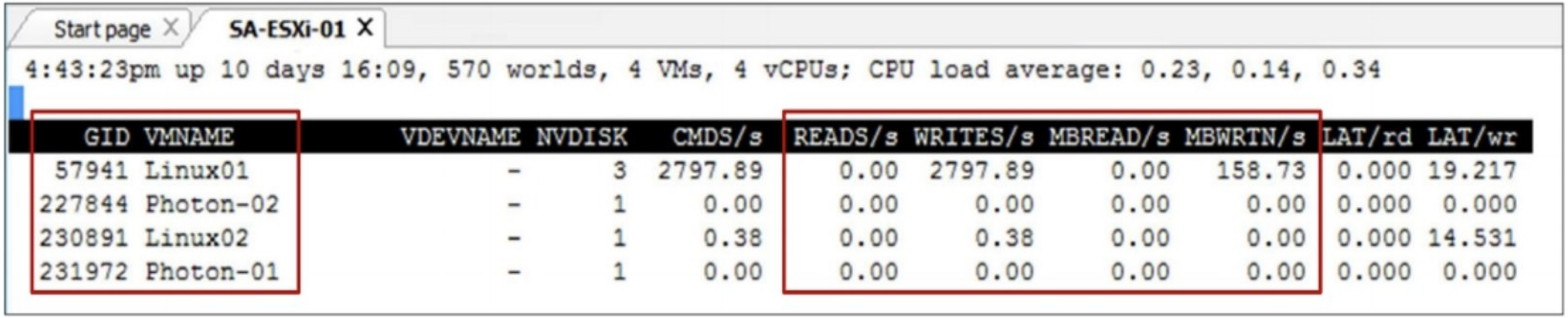
• MBWRTN/s: Number of megabytes written per second

All of these metrics can be monitored per HBA (vmhba#).

- The esxtop screen on the slide shows disk activity per disk adapter. To display the set of fields, enter f. Ensure that the following fields are selected: A (adapter name), B (path name), C (number of paths), E (I/O statistics), and G (overall latency statistics). If necessary, select fields to display by entering a, b, c, e, or g.

Monitoring Individual VMDK Statistics for a VM

On the `esxtop` screen, enter **v** to display disk throughput by virtual machine.



Start page
SA-ESXi-01

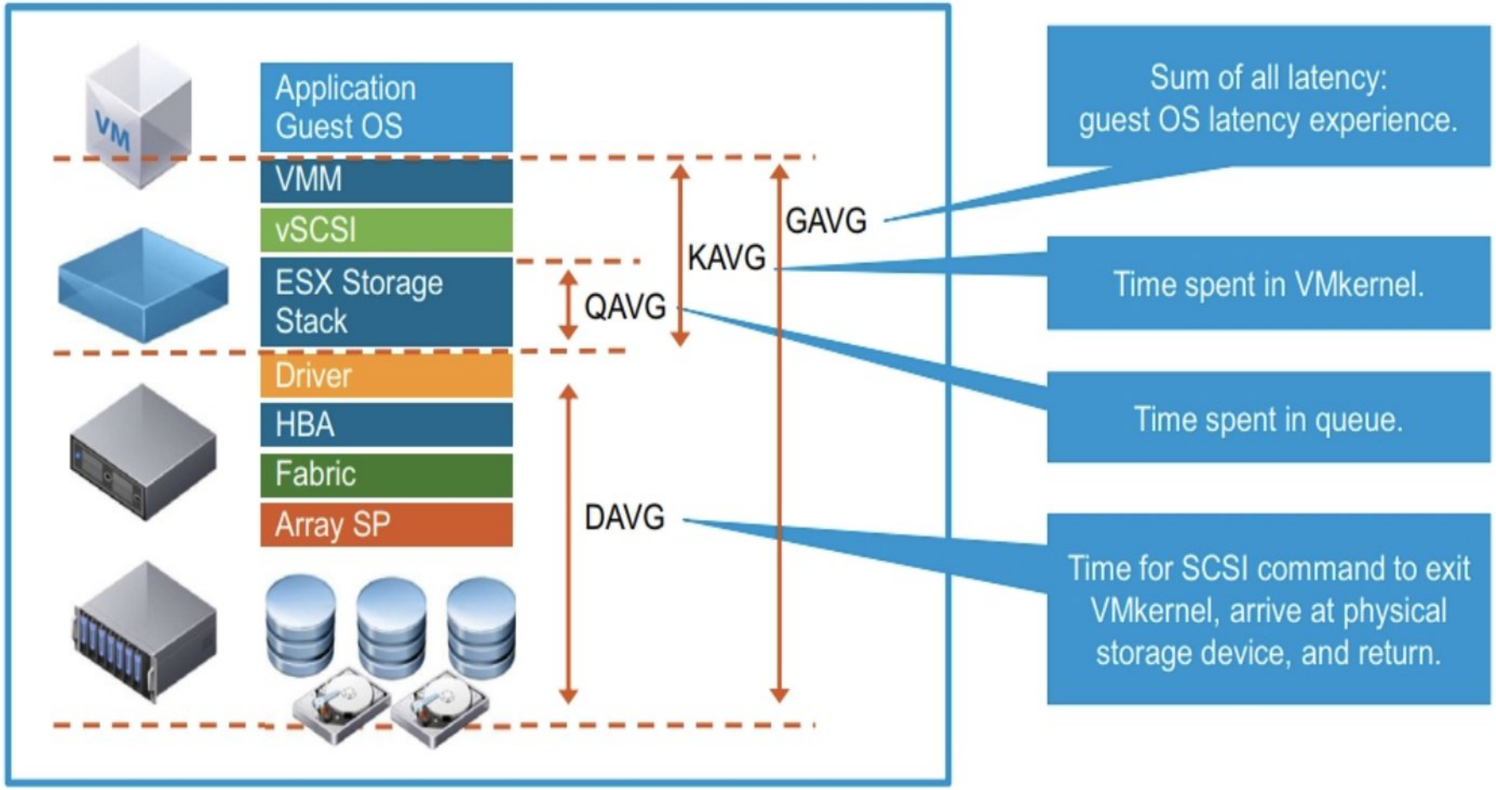
4:43:23pm up 10 days 16:09, 570 worlds, 4 VMs, 4 vCPUs; CPU load average: 0.23, 0.14, 0.34

GID	VMNAME	VDEVNAME	NVDISK	CMDS/s	READS/s	WRITES/s	MBREAD/s	MBWRTN/s	LAT/rd	LAT/wr
57941	Linux01	-	3	2797.89	0.00	2797.89	0.00	158.73	0.000	19.217
227844	Photon-02	-	1	0.00	0.00	0.00	0.00	0.00	0.000	0.000
230891	Linux02	-	1	0.38	0.00	0.38	0.00	0.00	0.000	14.531
231972	Photon-01	-	1	0.00	0.00	0.00	0.00	0.00	0.000	0.000

Disk Latency Metrics

Disk latency is the time (measured in milliseconds) that a SCSI command spends in transit, from the source to the destination and back.

The following disk latency values can be monitored with `esxtop`.



Disk Latency Metrics

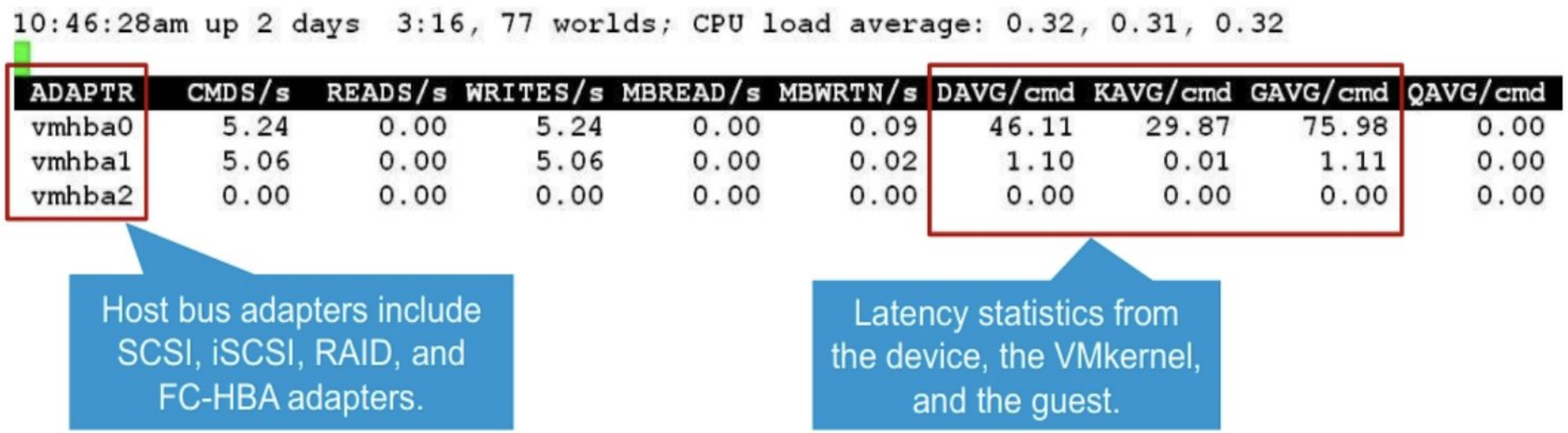
- Latency statistics are measured at the different layers of the ESXi storage stack.
- **GAVG** is the round-trip latency that the guest operating system sees for all I/O requests sent to the virtual storage device.
- **KAVG** tracks the latencies due to the ESXi VMkernel commands. The KAVG value should be very small in comparison to the DAVG value and it should be close to zero. When a lot of queuing occurs in ESXi, KAVG can be as high as or higher than DAVG. If this situation occurs, check the queue statistics.
- **DAVG** is the latency seen at the device driver level. It includes the round-trip time between the HBA and the storage. DAVG is a good indicator of performance of the back-end storage. If I/O latencies are suspected of causing performance problems, DAVG should be examined.
- **QAVG** is the average queue latency. QAVG is part of KAVG. Response time is the sum of the time spent in queues in the storage stack and the service time spent by each resource in servicing the request. The largest component of the service time is the time spent retrieving data from physical storage. If QAVG is high, consider examining the queue depths at each level in the storage stack.

continue

- **ADAPTR** is the name of the host bus adapter (vmhba#), which includes SCSI, iSCSI, RAID, Fibre Channel, and FCoE adapters.
- **DAVG/cmd** is the average amount of time it takes a device, which includes the HBA, the storage array, and everything in between, to service a single I/O request (read or write). A value of less than 10 indicates that the system is healthy. A value in the range of 11 through 20 indicates that you must monitor the value more frequently. A value greater than 20 indicates a problem.
- **KAVG/cmd** is the average amount of time it takes the VMkernel to service a disk operation. This number represents the time taken by the CPU to manage I/O. Because processors are much faster than disks, this value should be close to zero. A value of 1 or 2 is considered high for this metric.
- **GAVG/cmd** is the total latency seen from the virtual machine when performing an I/O request. GAVG is the sum of DAVG plus KAVG.

Monitoring Disk Latency with esxtop

In addition to disk throughput, the `esxtop` disk adapter screen (enter **d** in the window) enables you to monitor disk latency.



10:46:28am up 2 days 3:16, 77 worlds; CPU load average: 0.32, 0.31, 0.32

ADAPTR	CMDS/s	READS/s	WRITES/s	MBREAD/s	MBWRTN/s	DAVG/cmd	KAVG/cmd	GAVG/cmd	QAVG/cmd
vmhba0	5.24	0.00	5.24	0.00	0.09	46.11	29.87	75.98	0.00
vmhba1	5.06	0.00	5.06	0.00	0.02	1.10	0.01	1.11	0.00
vmhba2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

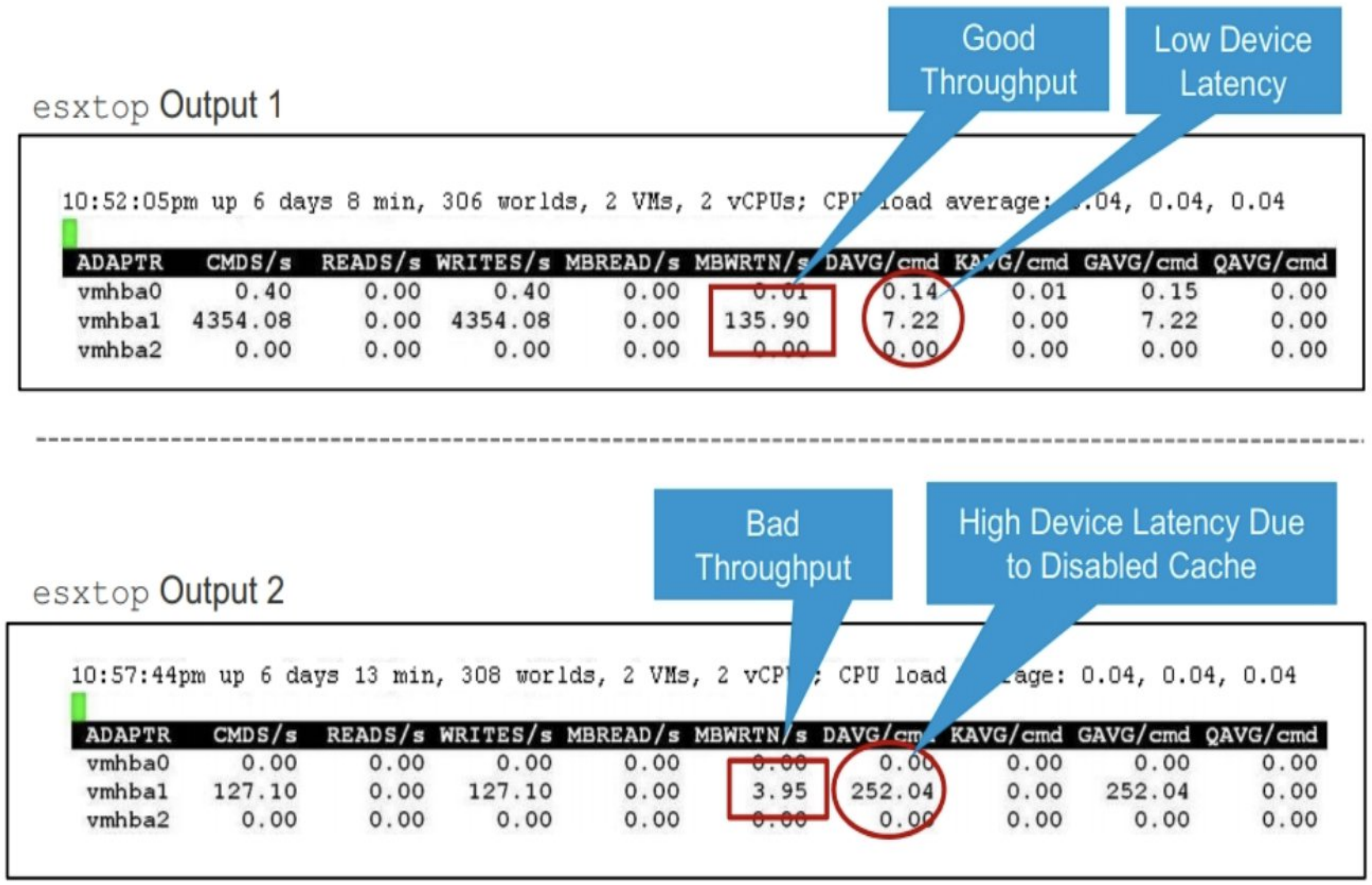
DAVG/cmd: Average latency (ms) of the device (LUN).

KAVG/cmd: Average latency (ms) in the VMkernel, also called queuing time.

GAVG/cmd: Average latency (ms) in the guest. GAVG = DAVG + KAVG.

Bad Disk Throughput

View latency values to determine if a performance problem truly exists.



Question:

What `esxtop` fields reflect the physical storage device's latency? Select all that apply.

- ❑ GAVG
- ❑ KAVG
- ❑ QAVG
- ❑ DAVG

Answer:

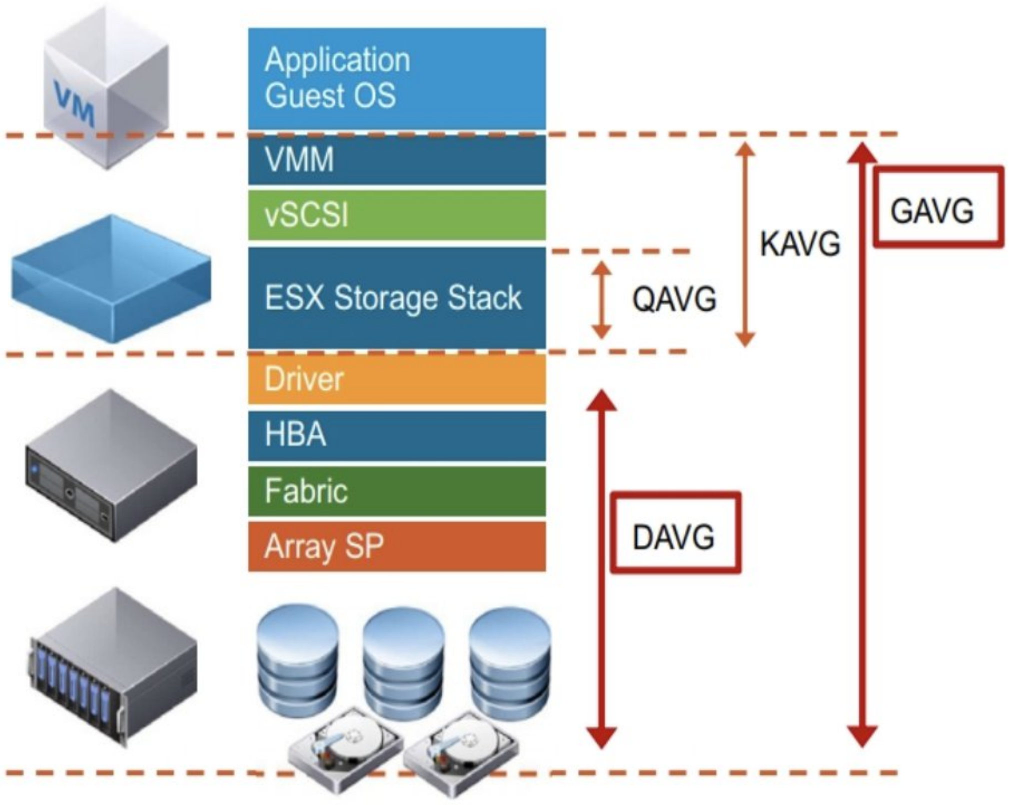
The following `esxtop` fields reflect a physical storage device's latency:

- ✓ GAVG
- ❑ KAVG
- ❑ QAVG
- ✓ DAVG

GAVG = KAVG + DAVG

DAVG is the latency seen at the device driver level.

The Array SP is directly affected by the latency of a physical storage device.



Overloaded Storage and Command Aborts

If a storage device is experiencing command aborts, the cause of these aborts must be identified and corrected.

To monitor command aborts in `esxtop`:

1. Enter **u** to display disks (LUNs).
2. Enter **f** and **L** to add error statistics to the display.

If ABRTS/s is more than 0 for any LUN, then storage is overloaded on that LUN.

Start page X | SA-ESXi-01 X

```
5:02:15pm up 1 day 18:14, 467 worlds, 1 VMs, 1 vCPUs; CPU load average: 0.21, 0.33, 0.37
```

DEVICE	FCMDS/s	FREAD/s	FWRITE/s	FMBRD/s	FMBWR/s	FRESV/s	ABRTS/s	RESETS/s
mpx.vmhba0:C0:T0:L0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
mpx.vmhba1:C0:T0:L0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
mpx.vmhba1:C0:T1:L0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
naa.60003ff44dc75adc8dcc0331f22a1c59	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Using esxtop to Identify Network Performance Problems

- When using esxtop or resxtop to troubleshoot a perceived networking issue, look out for dropped packets. Dropped transmit or received packets indicate that either the destination for the packets is too busy to receive them, or the network is too busy to properly send them.

- Firstly the end device/VM will attempt to buffer the traffic until it can be processed. If this queue fills, then the traffic is queued on the vSwitch. Finally, if the vSwitch queue is exhausted, then the packets will be dropped. You can check for dropped packets by looking at the **%DRPTX** and **%DRPRX** metrics in esxtop:

esxtop to Identify Network Performance Problems

1:49:01pm up 5:12, 290 worlds, 0 VMs, 0 vCPUs; CPU load average: 0.01, 0.01, 0.01

PORT-ID	USED-BY	TEAM-PNIC	DNAME	PKTTX/s	MbTX/s	PKTRX/s	MbRX/s	%DRPTX	%DRPRX
16777217	Management	n/a	vSwitch0	0.00	0.00	0.00	0.00	0.00	0.00
16777218	vmnic0	-	vSwitch0	4.41	0.26	8.64	0.01	0.00	0.00
16777219	vmnic1	-	vSwitch0	0.00	0.00	0.00	0.00	0.00	0.00
16777220	vmk0	vmnic0	vSwitch0	4.41	0.26	4.24	0.00	0.00	0.00
16777221	vmk3	vmnic1	vSwitch0	0.00	0.00	0.17	0.00	0.00	0.00
33554433	Management	n/a	vSwitch1	0.00	0.00	0.00	0.00	0.00	0.00
33554434	vmnic4	-	vSwitch1	0.00	0.00	0.00	0.00	0.00	0.00
33554435	vmnic5	-	vSwitch1	0.00	0.00	0.00	0.00	0.00	0.00
33554436	vmk1	vmnic4	vSwitch1	0.00	0.00	0.00	0.00	0.00	0.00
50331649	Management	n/a	vSwitch3	0.00	0.00	0.00	0.00	0.00	0.00
50331650	vmnic6	-	vSwitch3	0.00	0.00	0.00	0.00	0.00	0.00
50331651	vmnic7	-	vSwitch3	0.00	0.00	0.00	0.00	0.00	0.00
67108865	Management	n/a	DvsPortset-0	0.00	0.00	0.00	0.00	0.00	0.00
67108866	vmnic8	-	DvsPortset-0	0.00	0.00	0.00	0.00	0.00	0.00
67108867	vmnic9	-	DvsPortset-0	0.00	0.00	0.00	0.00	0.00	0.00

esxtop Network parameters:

- Dropped packets can occur for a number of reasons. Possible causes include high guest CPU utilisation, having incorrect network drivers installed in the guest OS. It could also be that there isn't sufficient uplink capacity in the vSwitch to cope with demand. If this is the case, consider migrating VMs to alternate hosts to reduce load, or add additional uplinks to the affected vSwitch.
- You can also look at implementing network I/O control, to help make better use of existing host resources. The other esxtop network metrics to be aware of are:
- MbTX/s – Amount of data transmitted in Mbps
- MbRX/s – Amount of data received in Mbps
- PKTTX/s – Average number of packets transmitted per second in the sampling interval
- PKTRX/s – Average number of packets received per second in the sampling interval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Resource

- https://kb.vmware.com/s/article/1008205
- http://www.yellow-bricks.com/esxtop/
- https://communities.vmware.com/docs/DOC-5420
- https://communities.vmware.com/docs/DOC-9279
- https://www.vmware.com/content/dam/digitalmarketing/vmware/en/pdf/techpaper/hpm-performance-vsphere55-white-paper.pdf